

Route-F P54: Covariant Lower Action, Finite Yukawa Operators, and Moving Seesaw Thresholds

P54PQ-v2 four-dimensional mainline

6 September 2026

Abstract

The previous common-phase/PS-flow checkpoint was committed and pushed as [4ad74e7](#). Keeping that same action fixed, we construct a local PS-covariant scalar/vector effective action and compute actual finite pure-Yukawa diagrams at the upper saddle. Split real six-plet masses generate conjugate-bidoublet operators outside the former four-matrix boundary. Their coefficients are calculable functions of the original two family matrices, not additional fitting parameters. Independently, we compute the full four-doublet scalar-cubic kinetic matrix and implement moving, possibly crossing or degenerate, sterile-neutrino thresholds. The latter includes the corrected dimension-five feedback found in 2024. These are bounded matching results, not a completed gauge/ghost finite matching calculation, physical flavor fit, or Higgs pole prediction.

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1 Frozen action and approximation ledger

The field content remains $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$ with the action and common phase dictionary in the preceding P54 cards. There are still only two symmetric UV family spurions $h = h_{\text{raw}}$ and $f = f_{\text{raw}}$. All scalar masses and vertices below come from that same 328-real-coordinate polynomial. No new vacuum or lattice scan, field multiplet, free mirror matrix, or physical scale is fitted here.

The actual broken background has $(\omega, \sigma, v_s) = (1, 0.1265, 0.25)$. The nearby PS saddle has $(1.00081030592151, 0, 0.254277697083708)$. Historical dimensionful unification scales remain unre-certified. A synthetic family benchmark demonstrates covariance and implementation, not agreement with measured masses or mixings.

There are three distinct approximations: (i) a tree local two-derivative PS action after upper-field elimination; (ii) finite one-loop diagram subsets with frozen tree propagators; (iii) one-loop dimension-five running with tree threshold matching. They cannot simply be summed into a claimed two-site one-loop prediction. The appropriate common hard-minus-EFT subtraction is still required.

2 A covariant lower action, not parent-weighted logarithms

2.1 Local scalar valley

Let B_A be the raw 248-real-dimensional PS parent embedding and B_H the 55-dimensional upper physical scalar embedding. In a local slice,

$$X(y) = x_0 + B_A y + B_H z(y), \quad B_H^T \nabla V(X(y)) = 0. \quad (1)$$

At $y = 0$, define $H = V^{(2)}(x_0)$ and

$$C = B_H^T H B_H, \quad W = B_H^T H B_A, \quad D = -C^{-1}W, \quad (2)$$

$$T = X_{,y} = B_A + B_H D, \quad X_{,ab} = -B_H C^{-1} B_H^T V^{(3)}[T_a, T_b]. \quad (3)$$

The measured $\lambda_{\min}(C) = 0.099450257411 \omega^2 > 0$ proves existence of an analytic valley germ by the implicit-function theorem. It proves neither global continuation nor uniform scale separation. Its local potential Hessian is $S = T^T H T$.

2.2 The 24-vector response

For the real generators $R_A = -iT_A$, the convention $D = \partial - igAT$ means $D = \partial + gAR$. Write $O_A = R_A X$, $I = O^T O$ and $j = gO^T D_{\text{PS}} X$. The scalar kinetic term is

$$\frac{1}{2} \|D_{\text{PS}} X + gOA_U\|^2. \quad (4)$$

Its algebraic heavy-vector solution and substitution are

$$A_U^* = -g^{-1} I^{-1} O^T D_{\text{PS}} X, \quad (5)$$

$$\Delta \mathcal{L}_2 = -\frac{1}{2} j^T (g^2 I)^{-1} j, \quad (6)$$

$$\mathcal{L}_{\text{EFT}}^{(2)} = \frac{1}{2} (Dy)^T g(y) (Dy) - V(X(y)), \quad g = T^T (1 - OI^{-1}O^T)T. \quad (7)$$

The 24 directions are a symmetric coset, not a gauge subgroup: $[\mathfrak{p}, \mathfrak{u}] \subset \mathfrak{u}$ and $[\mathfrak{u}, \mathfrak{u}] \subset \mathfrak{p} \neq 0$. Thus $I^{-1}O^T T$ is a PS-covariant response one-form, not a principal connection for an invented 24-dimensional group. The axion is held fixed as a gauge-neutral spectator in this chart.

For a direction q , with $O_q = (R_A X_q)_A$,

$$I_q = O_q^T O + O^T O_q, \quad K = O^T T, \quad K_q = O_q^T T + O^T T_q, \quad (8)$$

$$a = I^{-1}K, \quad a_q = I^{-1}(K_q - I_q a), \quad (9)$$

$$g_q = T_q^T T + T^T T_q - K_q^T a - K^T a_q. \quad (10)$$

These supply actual derivative vertices, not merely a mass census. The previous horizontal Schur metric is reproduced to 1.1×10^{-15} ; its eigenvalue interval is $[0.984249788817, 1.01800727050]$. All nine lower gauge masses and Goldstone normalizations agree with the full action. The current-current subtraction has rank 12 and trace 0.189002534197.

2.3 Nonlinear vertices and limits of the derivative expansion

For the exported corrected neutral ray q , put $J_H = B_H^T V^{(3)}[q, q]$. With $H^0 = h/\sqrt{2}$,

$$\lambda_{\text{upper}} = \frac{1}{6} V^{(4)}[q^4] - \frac{1}{2} J_H^T C^{-1} J_H = 0.407929900088, \quad \lambda_{\text{direct}} = 0.427823639280. \quad (11)$$

The induced derivative term is $0.024787549073 h^2 (\partial h)^2 / \omega^2$. These are the projected tree potential and metric on a bosonic-improved ray, not an SM quartic or a pole mass. The retained, canonically normalized 126 triplet has $m^2 = 0.268832826146 \omega^2$ and hh source 0.00252743109350ω ; eliminating it reconstructs the full mixed 54/126 tree response.

Already at quadratic order a heavy scalar gives the exact Euclidean kernel, in a raw scalar chart,

$$K_E(p^2) = p^2 G_0 + A - W^T (C + p^2 I)^{-1} W = S + p^2 G - p^4 W^T C^{-3} W + \dots \quad (12)$$

The corresponding exact pole calculation uses analytic continuation, not a replacement of all generalized eigenvalues by exact masses. The Yang–Mills term likewise yields higher-derivative operators upon substituting A_U^* into F_U and F_{PS} . Only modes coupled through W need this resolvent test; comparing every retained mass to a single global minimum heavy mass would impose an unnecessary restriction. The public API retains the matrices needed to handle the genuinely coupled modes exactly. For the twelve near-crossing modes, a source-selected invariant block is

$$H_2/\omega^2 = \begin{pmatrix} .2688378 & .0013417351173 \\ .0013417351173 & .261600225 \end{pmatrix}. \quad (13)$$

Its exact eigenvalues are .261359494901 and .269078530099. This is a finite resolvent repair, not a reason to redesign the action.

3 Actual finite upper Yukawa diagrams

3.1 Kernels, signs and scale consistency

Use $\mathcal{L}_Y = -Y_{IJ}^a \psi_I \psi_J s_a / 2 + \text{h.c.}$ with real canonical scalars and symmetric Weyl matrices. At the PS saddle all 48 Weyl fermions are massless. For a positive heavy scalar mass M_b^2 define

$$f_b = -\frac{1}{16\pi^2} \int_0^1 \log \frac{(1-t)M_b^2}{\mu^2} dt = \frac{1 - \log(M_b^2/\mu^2)}{16\pi^2}, \quad (14)$$

$$h_b = \frac{1}{16\pi^2} \int_0^1 (1-t) \log \frac{(1-t)M_b^2}{\mu^2} dt = \frac{\frac{1}{2} \log(M_b^2/\mu^2) - \frac{1}{4}}{16\pi^2}. \quad (15)$$

The fermion self-energy shift supplies the factor $(1-t)$; the two massless Weyl numerators in the triangle cancel one denominator. Differentiating the two fermions cancels the action's symmetry factor. The finite result, with $Z_\psi = 1 + K_\psi$, is

$$K_\psi = - \sum_{b \in H} Y_b^\dagger Y_b h_b, \quad (16)$$

$$\delta Y_a^{1\text{PI}} = - \sum_{b \in H} Y_b Y_a^\dagger Y_b f_b, \quad (17)$$

$$\Delta Y_a = \delta Y_a^{1\text{PI}} - \frac{1}{2} (K_\psi^T Y_a + Y_a K_\psi). \quad (18)$$

The tensor normalization is computed from the actual Clifford intertwiners, not imported from a different GUT. As an independent test,

$$\frac{d\Delta Y_a}{d\log\mu} = -\frac{1}{16\pi^2} \left[\frac{1}{2}(S_H^T Y_a + Y_a S_H) + 2 \sum_b Y_b Y_a^\dagger Y_b \right], \quad S_H = \sum_b Y_b^\dagger Y_b. \quad (19)$$

This equals minus the actual UV-minus-PS beta contraction. All-massless fermion contributions to scalar wave functions cancel in the upper hard subtraction. A scalar-cubic/two-Yukawa triangle has no dimension-four zero-momentum contribution with a massless internal fermion; it requires a chirality flip or external momentum. This does not eliminate scalar cubic wave-function diagrams.

3.2 Finite conjugate-bidoublet operators

Only 24 of the 55 heavy real scalars have nonzero Yukawa couplings:

Parent	Squared masses / ω^2	Multiplicities
10 six	0.190903147974, 0.200245831815	6, 6
126 six	0.372275090654, 0.415462413403	6, 6

For a complete degenerate pair $Y_I = \pm i Y_R$, $Y_R Y_a^\dagger Y_R + Y_I Y_a^\dagger Y_I = 0$. Unequal real masses weight these terms differently. Hence this cancellation fails by an exact split logarithm, not roundoff.

Write $\Delta_x = \log(m_{x,+}^2/m_{x,-}^2)/(16\pi^2)$. In the exported normalized mirror-intertwiner phase convention,

$$\tilde{H} = i(3\sqrt{2}\Delta_h h h^\dagger h - 6\sqrt{2}\Delta_f f h^\dagger f), \quad (20)$$

$$\tilde{F} = i(-4\Delta_h h f^\dagger h + 8\Delta_f f f^\dagger f). \quad (21)$$

Their four real coefficients, in the displayed order, are

$$(0.00128368533616, -0.00589775824991, -0.00121027014148, 0.00556045980308). \quad (22)$$

The fermion metric is $K_\psi = I_{16} \otimes k$, where

$$k = 0.0186815380325 h^\dagger h + 0.0108097087894 f^\dagger f \quad (23)$$

at $\mu/\omega = 0.563058770439$.

The holomorphic projection still satisfies $F = \sqrt{2}R$ and $L = R$ in this subset. But it is not the whole finite boundary. The mirror matrices are orthogonal intertwiner directions and must be retained. They are nonlinear functions of the same h, f , not new UV parameters. The displayed i is a tensor-basis phase. Restoring the PQ orbit gives the required $e^{-4i\theta}$ spurion phase; there is no explicit PQ breaking. Mirror terms vanish in the degenerate limit and are matching scale independent.

A strict one-loop matching-plus-running expansion need not feed an already one-loop mirror threshold back into a one-loop beta function: that effect starts at two loops. A resummed flow retaining mirror matrices must, however, evolve the complete real-scalar Weyl tensor. It cannot use the former four-holomorphic-parent equations as a closed system. In particular the paired-holomorphic fast cancellation cannot be assumed for a scalar with both orientations of Yukawa coupling.

The six invariant families $H, F, L, R, \tilde{H}, \tilde{F}$ do close under the generic one-loop real-Weyl contraction. The four bidoublet matrices are general 3×3 matrices; L, R are symmetric, giving $4 \cdot 9 + 2 \cdot 6 = 48$ complex tensor components, not 48 new UV parameters. PS representation decomposition excludes further dimension-four Yukawa tensors. Both generic six-family and actual finite-boundary tests reconstruct the complete beta tensor with relative error below 5×10^{-15} . Restriction to four families

leaves a nonzero 3.09×10^{-4} beta-tensor norm in the actual diagnostic. For reproducibility the evolved object is the symmetric real-scalar Weyl tensor Y_a , with $S_Y = \sum_b Y_b^\dagger Y_b$:

$$\begin{aligned} \mathcal{B}_{Y_a} = & \frac{1}{2}(S_Y^T Y_a + Y_a S_Y) + 2 \sum_b Y_b Y_a^\dagger Y_b \\ & + \sum_b Y_b \operatorname{Re} \operatorname{Tr}(Y_b^\dagger Y_a) - 3 \sum_i g_i^2 (C_i^T Y_a + Y_a C_i). \end{aligned} \quad (24)$$

Projecting (24) onto all six actual intertwiners defines the closed family flow without presuming a holomorphic-pair cancellation. The scalar and Weyl ordering is unchanged from the 248-real-parent and 48-Weyl common basis.

4 A genuine momentum-dependent scalar matching subset

For $V \supset \eta_i(M_{ij}^2 + T_{aij}q_a)\eta_j/2$, the real-scalar functional determinant yields

$$\Gamma_1 = \frac{1}{2} \operatorname{Tr} \log(K_0 + Tq + \dots), \quad \Gamma_{1,qq} = -\frac{1}{4} \operatorname{Tr}(GTqGTq) + \dots \quad (25)$$

The UV-finite Euclidean two-point difference is

$$\begin{aligned} \Pi_{ab}(p_E^2) - \Pi_{ab}(0) = & \frac{1}{32\pi^2} \sum_{ij}^{\text{at least one hard}} T_{aij} T_{bji} \\ & \times \int_0^1 \log \left(1 + \frac{t(1-t)p_E^2}{(1-t)m_i^2 + tm_j^2} \right) dt. \end{aligned} \quad (26)$$

Both hard-hard and hard-soft pairs are required. Purely soft pairs belong to the EFT and are subtracted. Differentiation gives

$$K_{ab}^S = \frac{1}{32\pi^2} \sum_{ij}^{\text{hard}} T_{aij} T_{bji} I(m_i^2, m_j^2), \quad (27)$$

$$I(x, y) = \frac{x^2 - y^2 - 2xy \log(x/y)}{2(x-y)^3}, \quad I(x, x) = \frac{1}{6x}, \quad I(x, 0) = \frac{1}{2x}. \quad (28)$$

This is positive semidefinite as a weighted Gram matrix. The single real-heavy test $ghS^2/2$ gives $K_h = g^2/(192\pi^2 M^2)$. Quartic tadpoles are momentum independent. One-loop mass counterterms inside this one-loop graph would be selected two-loop contributions. The massless Goldstone propagators inherit background-field Landau $\xi = 0$; this scalar subset is not separately gauge independent. With exactly zero soft masses, mixed-bubble soft-region momentum expansions are scaleless in dimensional regularization. Finite soft masses or a staged subtraction require a fresh EFT calculation.

All 16 real components of the four doublets are assembled using the actual scalar Hessian jets and electroweak rotations. The result obeys all four electroweak Ward identities. Its neutral complex matrix is

$$K_4^S \simeq \begin{pmatrix} .00206838 & -.00033316 & -.00000208 & -.00009074 \\ -.00033316 & .00213591 & -.00001766 & .00000379 \\ -.00000208 & -.00001766 & .07100660 & -.00045735 \\ -.00009074 & .00000379 & -.00045735 & .08599460 \end{pmatrix}, \quad (29)$$

with imaginary norm 3.5×10^{-18} , checked rather than imposed. On the bosonic-improved light vector,

$$\delta Z_h = c_B^\dagger K_4^S c_B = 0.00243676715822. \quad (30)$$

The covariant quadratic operator is $(D_\mu H)^\dagger K_4^S (D^\mu H)$. Gauge vertices required by this operator are not a substitute for independent gauge-loop, ghost and Nielsen calculations.

For $Z = 1 + K_4^S$, the canonical operation is

$$D_c = Z^{-1/2} D Z^{-1/2}, \quad c_c = \frac{Z^{1/2} c}{\sqrt{c^\dagger Z c}}, \quad Y_c(c_c) = \frac{Y(c)}{\sqrt{c^\dagger Z c}}. \quad (31)$$

The down/lepton expressions use conjugate scalar factors. The zero eigenvector survives by congruence; its copy-coordinate changes are not an additional physical light-state rotation. The scalar-leg-only factor is 0.998783838597. Exact square roots verify the algebra, not a two-loop resummation claim. No 55-upper contribution is added again to this one-step 290-hard calculation.

5 Mass-ordered seesaw as a changing effective theory

5.1 Conventions and running

Let $H^0 = v/\sqrt{2}$, $V = -m^2 H^\dagger H + \lambda(H^\dagger H)^2$ and

$$M_\nu = \frac{v^2}{2}(C_5 - Q), \quad Q = Y_\nu M_R^{-1} Y_\nu^T. \quad (32)$$

Store $\widehat{M} = M_R/\omega$, $\widehat{C}_5 = \omega C_5$. The actual local type-II input obeys $\widehat{C}_5^{II} = 2C_{II,\text{raw}} f_{\text{raw}}$; this factor is tested against the previous full local I+II matrix. The diagnostic boundary $g = (.53, .55, .60)$, $\lambda = .12$, $\mu_{\text{start}}/\omega = .1$ is explicitly not a solved P54 matching condition.

Define $H_x = Y_x Y_x^\dagger$, $S_N = Y_\nu^\dagger Y_\nu$, $T = \text{Tr}(3H_u + 3H_d + H_e + H_\nu)$, $T_4 = \text{Tr}(3H_u^2 + 3H_d^2 + H_e^2 + H_\nu^2)$ and $\mathcal{B} = 16\pi^2 d/d \log \mu$. In GUT hypercharge normalization, the $C_{HN} = C_{BN} = 0$ subsystem is

$$\mathcal{B}_{Y_u} = [\tfrac{3}{2}(H_u - H_d) + T - \tfrac{17}{20}g_1^2 - \tfrac{9}{4}g_2^2 - 8g_3^2]Y_u, \quad (33)$$

$$\mathcal{B}_{Y_d} = [\tfrac{3}{2}(H_d - H_u) + T - \tfrac{1}{4}g_1^2 - \tfrac{9}{4}g_2^2 - 8g_3^2]Y_d, \quad (34)$$

$$\mathcal{B}_{Y_e} = [\tfrac{3}{2}(H_e - H_\nu) + T - \tfrac{9}{4}g_1^2 - \tfrac{9}{4}g_2^2]Y_e, \quad (35)$$

$$\mathcal{B}_{Y_\nu} = [\tfrac{3}{2}(H_\nu - H_e) + T - \tfrac{9}{20}g_1^2 - \tfrac{9}{4}g_2^2]Y_\nu + 3C_5 Y_\nu^* M_R, \quad (36)$$

$$\mathcal{B}_{M_R} = S_N^T M_R + M_R S_N, \quad \mathcal{B}_{g_i} = b_i g_i^3, \quad b = (41/10, -19/6, -7), \quad (37)$$

$$\mathcal{B}_{C_5} = P_C C_5 + C_5 P_C^T + \alpha_C C_5, \quad (38)$$

$$P_C = -\tfrac{3}{2}H_e + \tfrac{7}{2}H_\nu, \quad \alpha_C = 4\lambda - 3g_2^2 + 2T, \quad (39)$$

$$\begin{aligned} \mathcal{B}_\lambda &= 24\lambda^2 - 3\lambda(3g_2^2 + \tfrac{3}{5}g_1^2) + \tfrac{3}{4}g_2^4 + \tfrac{3}{8}(g_2^2 + \tfrac{3}{5}g_1^2)^2 \\ &\quad + 4\lambda T - 2T_4 - 4 \text{Re Tr}(C_5 Y_\nu^* M_R Y_\nu^\dagger). \end{aligned} \quad (40)$$

The 7/2, C_5 contribution to $\mathcal{B}_{Y_\nu}$ and the last quartic term include the 2024 correction [1]. Our C_5 is the negative of that reference's coefficient, accounting for the two signs. The former 1/2 equations are retained only as a negative regression. This distinction changes tensor trajectories even when its numerical effect in one chosen benchmark is small.

By the inverse-matrix product rule, with $P = -3H_e/2 + H_\nu/2$,

$$\mathcal{B}_Q = PQ + QP^T + \alpha_Q Q + 3(H_\nu C_5 + C_5 H_\nu^T), \quad (41)$$

$$\alpha_Q = 2T - \tfrac{9}{10}g_1^2 - \tfrac{9}{2}g_2^2, \quad (42)$$

$$\mathcal{B}_{C_5-Q} = P(C_5 - Q) + (C_5 - Q)P^T + \alpha_C(C_5 - Q) + (4\lambda + \tfrac{9}{10}g_1^2 + \tfrac{3}{2}g_2^2)Q. \quad (43)$$

The new terms cancel instantaneously in the difference, but change Y_ν and λ and thus its later trajectory. Evolving one initial Weinberg sum above all thresholds would omit the source in (43). The separate II tag is a source decomposition along one common trajectory, not a new independently fitted matrix.

5.2 Exact block-tree matching and live events

At a sterile split let

$$M' = U^T M_R U = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}, \quad Y' = Y_\nu U = (Y_r, Y_h). \quad (44)$$

Eliminating $N_h = -D^{-1}(B^T N_r + Y_h^T \ell H)$ gives

$$\boxed{C_5^- = C_5^+ - Y_h D^{-1} Y_h^T, \quad Y_r^- = Y_r - Y_h D^{-1} B^T, \quad M_r^- = A - B D^{-1} B^T.} \quad (45)$$

The block inverse identity proves continuity of $C_5 - Q$ and associativity of sequential Schur complements. This is exact matrix algebra at leading seesaw order, not exact finite- v diagonalization. For a genuinely mixed non-Takagi block, substituting the heavy solution also induces a retained kinetic metric; the displayed algebra is not by itself a canonical EFT matching prescription. Physical threshold events here use a Takagi basis, where $B = 0$ and this issue is absent.

The next event solves

$$t - \log \sigma_{\max}[\widehat{M}_R(t)] = 0, \quad t = \log(\mu/\omega), \quad (46)$$

with a freshly evaluated Takagi spectrum at each event call. An entire degenerate cluster is removed at once. Neither frozen eigenvalues nor a fixed initial ordering defines the event. Tests include complex family covariance, an actual running level crossing, non-diagonal Schur matching, a degenerate three-state block and a minimal two-sterile massless-light-neutrino regression.

The two initial local-light examples have moving event scales (.04648960, .02861195, .01786944) and (.06080995, .03932527, .02503881) in ω units, respectively, in the displayed zero- C_{HN} diagnostic subsystem. Running continues after each event and below the last one. These are not certified physical seesaw scales.

5.3 The operator-completeness question

The dimension-five sterile EFT also allows

$$\mathcal{O}_{HN} = \overline{N^c} N H^\dagger H, \quad \mathcal{O}_{BN} = \overline{N^c} i \sigma^{\mu\nu} N B_{\mu\nu}. \quad (47)$$

The zero- C_{HN}, C_{BN} subspace is invariant under the one-loop equations, but this does not prove the P54 scalar matching lies in it. For a heavy real neutral scalar vector s , write

$$V = \frac{1}{2} s^T K s + s^T J H^\dagger H + \dots, \quad \mathcal{L}_Y \supset -\frac{1}{2} N^T G_A s_A N + \text{h.c.} \quad (48)$$

Its tree equation $s = -K^{-1} J H^\dagger H + \dots$ gives, in the operator convention with no 1/2 in $C_{HN} \mathcal{O}_{HN}$,

$$C_{HN} = \frac{1}{2} \sum_A G_A (K^{-1} J)_A, \quad \frac{\partial M_R(H)}{\partial (H^\dagger H)} = -2C_{HN}. \quad (49)$$

Thus it must be computed from the actual neutral scalar source and Majorana intertwiner. Gauge and PQ zero modes must not be inverted. This is an operator-basis issue; declaring the coefficient small or zero without its matching calculation would not repair it.

That calculation is now performed on the exact tree light ray. The five-real-dimensional SM-singlet scalar plane contains three positive radial modes and two gauge/PQ zero modes. Inverting only the positive block in (49) gives

$$\boxed{\omega C_{HN} = -0.0848692011507 i f_{\text{raw}} \neq 0.} \quad (50)$$

The actual common-phase Majorana intertwiner is used, and its vacuum contraction reproduces $M_R = i4\sqrt{2}\sigma f_{\text{raw}}$. An independent check restricts the original potential to the three radial modes, solves its stationary equations at three nonzero Higgs fields and extracts $-\Delta M_R/h^2$. It converges to the displayed coefficient. This is not an assumption inferred from allowed operators. On the bosonic-improved ray the frozen-tree projection is $-0.252600617 i f_{\text{raw}}/\omega$; the two local-light projections are $-0.255045272 i f_{\text{raw}}/\omega$ and $-0.256536567 i f_{\text{raw}}/\omega$. Those are explicitly mixed-order projections, not loop-complete Wilson coefficients. The exact-tree result already disproves a zero- C_{HN} P54 boundary.

5.4 Required nonzero- C_{HN} evolution and threshold transport

Keep $C = C_{HN}$, $C_{BN} = 0$, and the already declared $V = -m^2 H^\dagger H + \lambda(H^\dagger H)^2$. The additional terms are

$$\Delta\mathcal{B}_{Y_\nu} = -4Y_\nu M_R^\dagger C, \quad \Delta\mathcal{B}_{M_R} = +8m^2 C, \quad (51)$$

$$\Delta\mathcal{B}_\lambda = 16 \operatorname{Re} \operatorname{Tr}(C^\dagger M_R S_N), \quad (52)$$

$$\mathcal{B}_C = a_H C + 4(C S_N + S_N^T C), \quad a_H = 2T - \frac{9}{10}g_1^2 - \frac{9}{2}g_2^2 + 12\lambda, \quad (53)$$

$$\mathcal{B}_{m^2} = a_H m^2 + 4 \operatorname{Tr}(M_R^\dagger M_R S_N) - 8 \operatorname{Re} \operatorname{Tr}(M_R^\dagger M_R M_R^\dagger C). \quad (54)$$

The sign of the last mass equations follows because the cited reference uses $V = +m_Z^2 H^\dagger H + \dots$, so $m_Z^2 = -m^2$ here. The implementation stores $q_H = m_Z^2/\omega^2 = -m^2/\omega^2$ directly, together with M_R/ω and ωC ; this explicit conversion avoids inferring a sign from the label ‘‘Higgs mass.’’ C_5 has no direct C mixing at this order. The zero dipole subsystem is one-loop invariant, and this nonderivative scalar tree exchange does not generate it. A finite loop dipole coefficient has not thereby been proven zero.

An independent normalization check uses one real family and $\mathcal{M}(h) = \begin{pmatrix} 0 & yh/\sqrt{2} \\ yh/\sqrt{2} & M-Ch^2 \end{pmatrix}$. The coefficient of h^4 in $\operatorname{Tr}[(\mathcal{M}^\dagger \mathcal{M})^2]$ is $y^4/2 - 4y^2 M C + O(C^2)$. The same Coleman–Weinberg divergence that gives $\mathcal{B}_\lambda = -2y^4$ therefore gives $+16y^2 M C$, independently checking the coefficient and sign in (52).

For a non-diagonal sterile split, expand $M_R(\rho) = M_R - 2C\rho$ with $\rho = H^\dagger H$ before taking its Schur complement. Differentiating once yields

$$C^- = C_{rr} - C_{rh} D^{-1} B^T - B D^{-1} C_{hr} + B D^{-1} C_{hh} D^{-1} B^T. \quad (55)$$

This reduces to the retained block in an exact Takagi basis. The ρ -dependent Yukawa shift and Weinberg denominator produce operators of dimension six and seven, respectively, outside this linear dimension-five truncation. They must not be mistaken for additional dimension-five matching terms.

Equation (43) is expressly the zero- C identity. With nonzero C , its additional source is minus

$$\Delta\mathcal{B}_Q = (\Delta\mathcal{B}_{Y_\nu}) M_R^{-1} Y_\nu^T + Y_\nu M_R^{-1} (\Delta\mathcal{B}_{Y_\nu})^T - Y_\nu M_R^{-1} (\Delta\mathcal{B}_{M_R}) M_R^{-1} Y_\nu^T. \quad (56)$$

Thus a full model run cannot continue to assert the zero- C sum-flow identity after the nonzero boundary is inserted. A diagnostic choice $m^2 = 0$ at one starting scale is not radiatively invariant and is not a physical electroweak matching condition. In the two nonzero- C diagnostic runs the terminal values are $q_H = 2.22133 \times 10^{-6}$ and 2.98359×10^{-6} , respectively. They exceed the squared terminal scale in ω units: a Higgs kept light all the way to that endpoint is not self-consistent without quadratic matching and retuning. The mass-independent $\overline{\text{MS}}$ ODE can be continued as a mathematical diagnostic, but its fixed- v terminal mass proxies are not a physically valid light-Higgs EFT endpoint.

6 Reproducibility and remaining physical work

The code and JSON/Markdown records under `route_f/code` and `route_f/output` are named

Stem after <code>verify_</code>	Computed scope
<code>p54_lower_covariant_eft</code>	scalar valley, vector response, local vertices
<code>p54_upper_yukawa_thresholds</code>	actual pure-Yukawa finite state sums
<code>p54_scalar_kinetic_matching</code>	full scalar-cubic four-doublet metric
<code>p54_scalar_chn</code>	actual neutral-scalar tree $NNH^\dagger H$ coefficient
<code>p54_sequential_seesaw</code>	moving thresholds and dimension-five flow

Each record declares its boundary data, source hashes, conventions, scope restrictions and executable positive and negative tests. Previously computed expensive Hessians are reused.

The next physical matching calculation must include the omitted heavy-vector fermion kinetic/vertex graphs, the vector/scalar/Goldstone/ ghost scalar kinetic package, upper scalar matching and tadpole responses, and the lower covariant hard-minus-EFT subtraction. Sequential finite one-loop type-I/type-II matching is separate from the tree Schur rule: an all-sterile formula cannot silently be reused when lighter sterile states and an existing type-II operator remain. Only after these matched inputs are assembled is a constrained global fit meaningful. Momentum-dependent pole self-energies and Nielsen consistency remain separate from a zero-momentum curvature condition. No soliton regularity result, determinant or portal is promoted here. The sterile running engine is a specified $C_{BN} = 0$ dimension-five subsystem with the PQ axion held as a background spectator. An invariant dipole-free RGE subspace is not a proof of zero finite dipole matching, and freezing the axion background is not a calculation of quantum axion loops. Its Wilson couplings and quantum effects must be accounted for before calling the result the full physical P54 lower evolution.

References

- [1] D. Zhang, *Threshold Effects on the Massless Neutrino in the Canonical Seesaw Mechanism* (2024), <https://arxiv.org/abs/2405.18017>.
- [2] S. Antusch et al., *Running Neutrino Mass Parameters in See-Saw Scenarios* (2005), <https://arxiv.org/abs/hep-ph/0501272>.
- [3] K. M. Patel and V. Shukla, finite Yukawa matching conventions and kernels, <https://arxiv.org/abs/2310.16563>.
- [4] M. Luo, H. Wang and Y. Xiao, general renormalization-group equations, <https://arxiv.org/abs/hep-ph/0211440>.
- [5] M. Gabelmann, M. Muehlleitner and F. Staub, *Automatised matching between two scalar sectors at the one-loop level*, <https://doi.org/10.1140/epjc/s10052-019-6570-5>.
- [6] D. Zhang and S. Zhou, complete one-loop type-I matching, <https://arxiv.org/abs/2107.12133>.